curriculum vitae

Computational Materials Scientist specializing in Integrated Computational Materials Engineering (ICME) and multi-scale materials modeling using machine-learning surrogate methods. Expertise in bridging microstructure to functional behavior for nuclear reactor materials and microelectronics applications. Combines deep computational mechanics with experimental characterization, including advanced microscopy (SEM, AFM, profilometry) and custom instrumentation design. Proven track record developing data-driven surrogate models from large simulation datasets, recognized with two R&D 100 Award Finalist selections.

Personalia

Surname:	Ruybalid
Given names	Andre, Paul
Degrees:	Ph.D. | MSc. | BSc.
City:	Los Alamos, New Mexico, U.S.A.
Phone number:	+1 719 210 1746 (personal mobile)
E-mail:	andreruybalid@gmail.com
Date of birth:	26 October 1987
Place of Birth:	Farmington, New Mexico, U.S.A.
Nationality:	American/Dutch

Professional experience

2023 - Present **R&D Engineer, Los Alamos National Laboratory**

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

- ❖ Developed multiple surrogate models using machine learning and data-driven methods to bridge polycrystalline mechanics of nuclear reactor materials to engineering-scale predictions from datasets curated from thousands of low-length-scale simulations.
- ❖ Coordinated with mechanistic modelers as part of multi-scale modeling workflow for metal alloy behavior under extreme conditions.
- ❖ Built and maintained material models for nuclear fuel performance analysis in MOOSE/BISON, including code contributions, verification, validation, and uncertainty quantification (UQ).
- ❖ Reached the R&D 100 Award Finals twice with surrogate modeling solutions for computational materials science applications.
- ❖ Collaborated with external industrial partners and national lab colleagues to deliver pragmatic modeling solutions; led technical milestone reports and peer-reviewed papers.

2021 – 2023 **Postdoctoral Researcher, Los Alamos National Laboratory**

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

Developed the surrogate model ("LAROMance") for polycrystalline nuclear reactor materials by designing large-scale simulation pipelines and curating datasets from thousands of low-length-scale simulations using machine learning approaches.

2019 – 2020 **Design Engineer, ASML**

DUV Def & Metrology group of Defectivity Department | Eindhoven, The Netherlands

Developed and validated advanced metrology systems for high-precision nanoscale defect detection on critical DUV lithography reticles, performing specialized cleanroom testing and data analysis under tight deadlines in collaboration with cross-functional and industrial teams.

2013 – 2019 **PhD Researcher, Eindhoven University of Technology**

Computational and Experimental Mechanics | Prof. Dr. Marc Geers & Dr. Johan Hoefnagels.
Philips Research, Materials Innovation Institute | Eindhoven, The Netherlands

Designed and implemented an integrated computational-experimental framework combining finite element modeling, custom instrumentation with piezoelectronic components and closed-loop feedback control, and microscale experimentation with advanced microscopy (SEM, AFM, profilometry) and sample preparation (sputter coating) to quantify adhesion in microelectronic systems (flexible OLEDs), resulting in five journal publications.

2017 – 2020 **Account manager & Technical recruiter**
Research-Square & EuFlex | Eindhoven, The Netherlands
Acquired and managed client accounts, and conducted technical recruitment of highly specialized engineering and PhD-level candidates for science and technology companies.

2018 – 2019 **Lab technologist, Multi-Scale Laboratory**
Eindhoven University of Technology | Eindhoven, The Netherlands
Computational and Experimental Mechanics
Managed an experimental mechanics laboratory, overseeing high-tech equipment including scanning electron microscopes and micro-mechanical test systems, while providing technical guidance to researchers and students and supporting experimental design in a high-demand environment.

2009 – 2013 **Tutor of students**
Eindhoven University of Technology, Mechanical Engineering Department | Freelance
Mentored bachelor student teams on engineering case studies, supporting balanced group dynamics, resolving team issues, and delivering constructive feedback.

Education

Eindhoven University of Technology, the Netherlands
Department of Mechanical Engineering
Division: Computational and Experimental Mechanics
Research Group: Mechanics of Materials

2013 – 2019 **PhD.-research project**
Title: "Identification of multi-layer interface systems: a full-field, micro-mechanical approach."
2013 – 2019 **Graduate School of Engineering Mechanics**
Advanced Mechanics Courses
2010 – 2013 **MSc. Mechanical Engineering**
Minor: Biomedical Engineering
2006 – 2010 **BSc. Mechanical Engineering**
2000 – 2006 **Research-oriented secondary education (VWO)**
St.-Janscollege, Hoensbroek, The Netherlands

Selected journal publications (peer-reviewed)

P. Robbe, A.P. Ruybalid, A. Hegde, C. Bonneville, H.N. Najm, L. Capolungo and C. Safta . A comparison of surrogate constitutive models for viscoplastic creep simulation of HT-9 steel. *Computational Materials Science*, 262, 114367 (2026). https://doi.org/10.1016/j.commatsci.2025.114367.

A.P. Ruybalid, A. Tallman, W. Wen, C. Matthews, L. Capolungo. Data-Driven Surrogate Modeling with Microstructure-Sensitivity of Viscoplastic Creep in Grade 91 Steel. Integrating Materials and Manufacturing Innovation 13, 895–914 (2024). https://doi.org/10.1007/s40192-024-00377-z

A.P. Ruybalid, O. van der Sluis, M.G.D. Geers and J.P.M. Hoefnagels. Full-field identification of mixed-mode adhesion properties in a flexible, multi-layer microelectronic material system. *Engineering Fracture Mechanics*, 226, pp. 106879 (2020), *https://doi.org/10.1016/j.engfracmech.2020.106879*.

A.P. Ruybalid, O. van der Sluis, M.G.D. Geers, J.P.M. Hoefnagels. An *In-Situ*, Micro-Mechanical Setup with Accurate, Tri-Axial, Piezoelectric Force Sensing and Positioning. *Exp Mech* 60, 713–725 (2020), *https://doi.org/10.1007/s11340-020-00583-8*.

A.P. Ruybalid, J.P.M. Hoefnagels, O. van der Sluis, M.P.F.H.L. van Maris and M.G.D. Geers. Mixed-mode cohesive zone parameters from integrated digital image correlation on micrographs only. *International Journal of Solids and Structures*, 156-157, pp. 179-193 (2019), *https://doi.org/10.1016/j.ijsolstr.2018.08.010*.